

Factors Influencing Rates of Metal Hydride Formation by Proton Transfer



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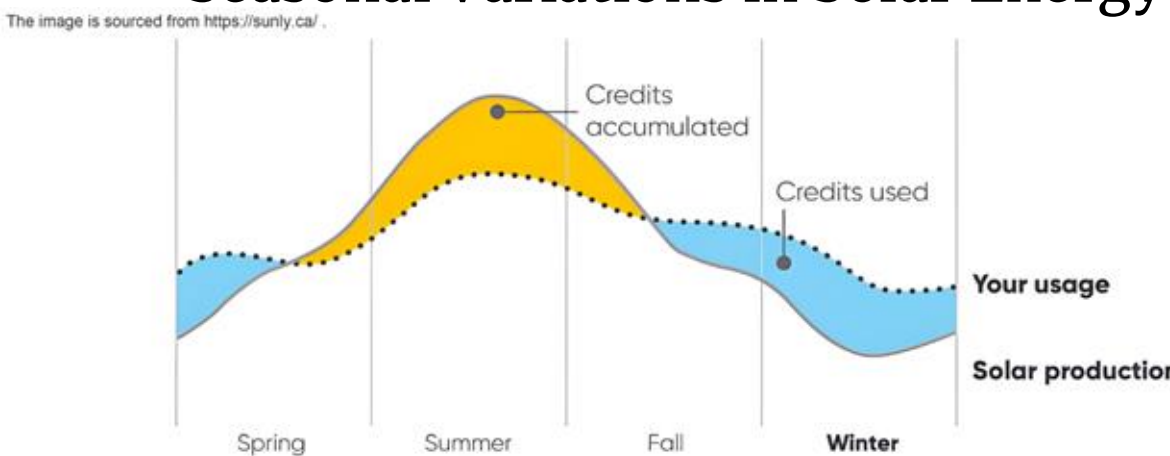


Introduction

Solar and Wind



Seasonal Variations in Solar Energy



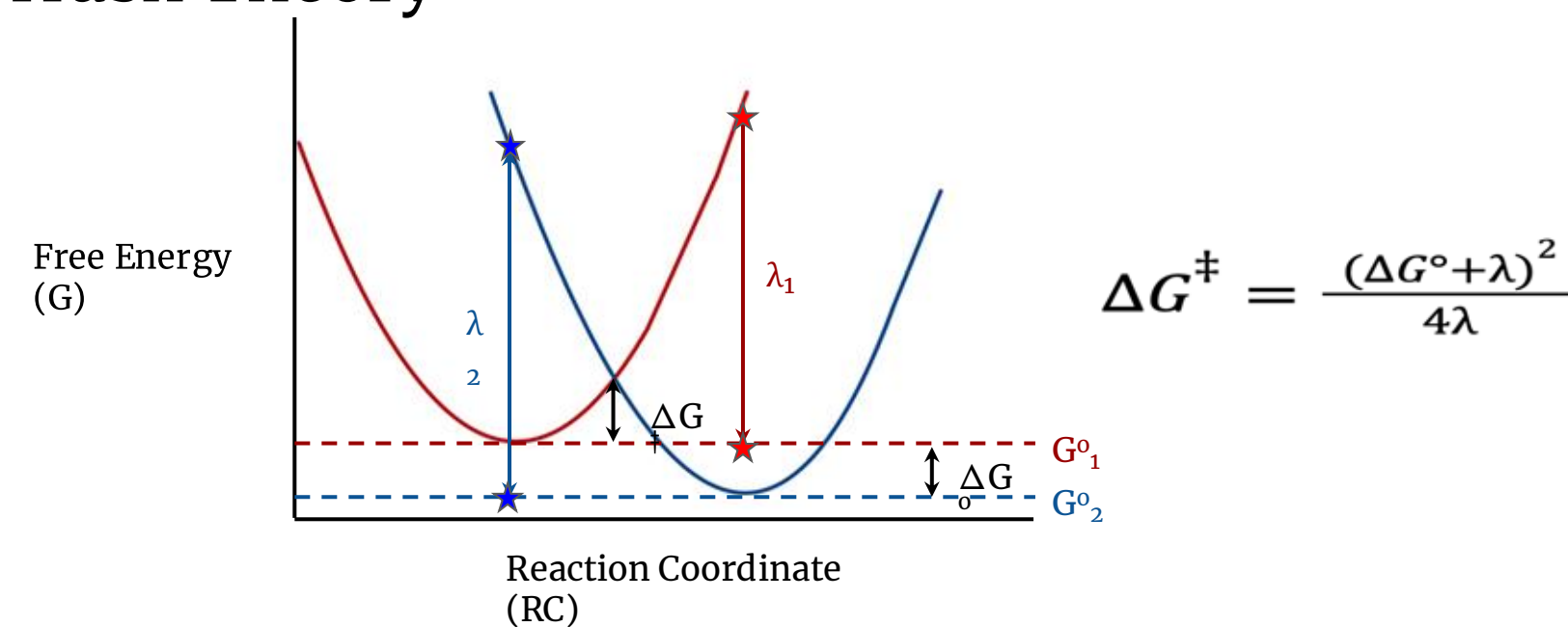
- Increased usage of renewable energy sources raises the need for energy storage options.
- Solar and wind power are clean, but long-term energy storage still proves to be unreliable.



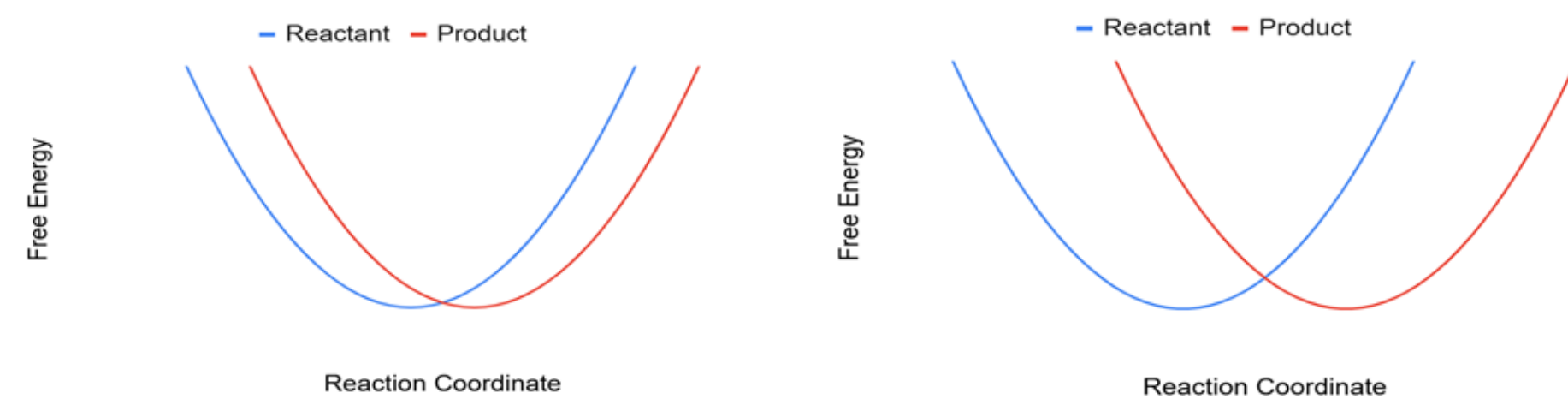
- Proton transfer-driven metal hydride formation is crucial to generating sustainable fuels.
- Reaction rate depends on many factors including reorganization energy and work terms.
- Understanding the rate of these PT reactions is crucial to improving overall efficiency of energy storage in chemical bonds.

Theoretical Background

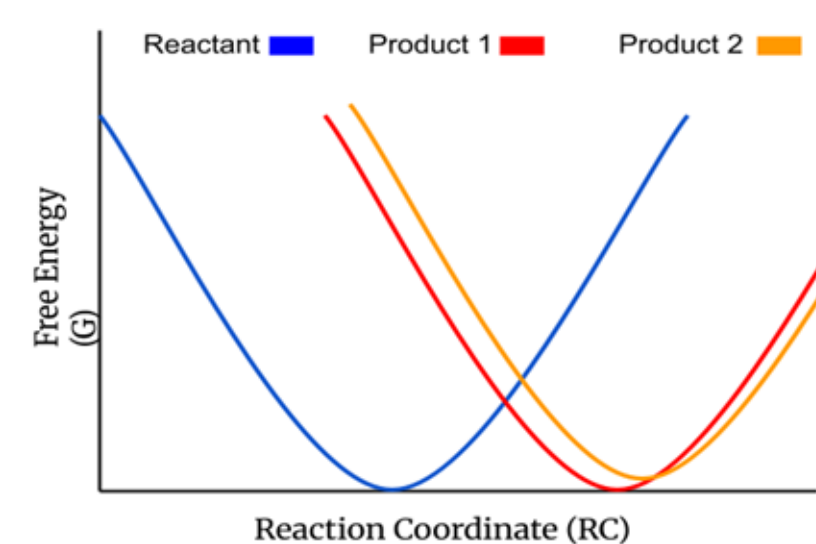
Marcus-Hush Theory



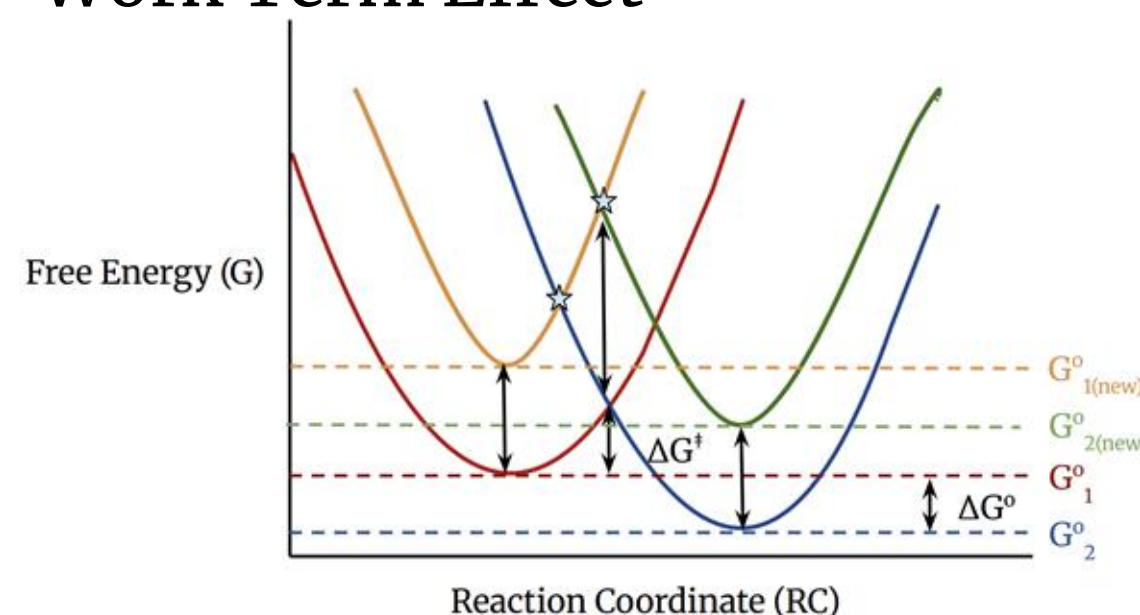
Reorganization Energy (λ) Effect



ΔG° Effect



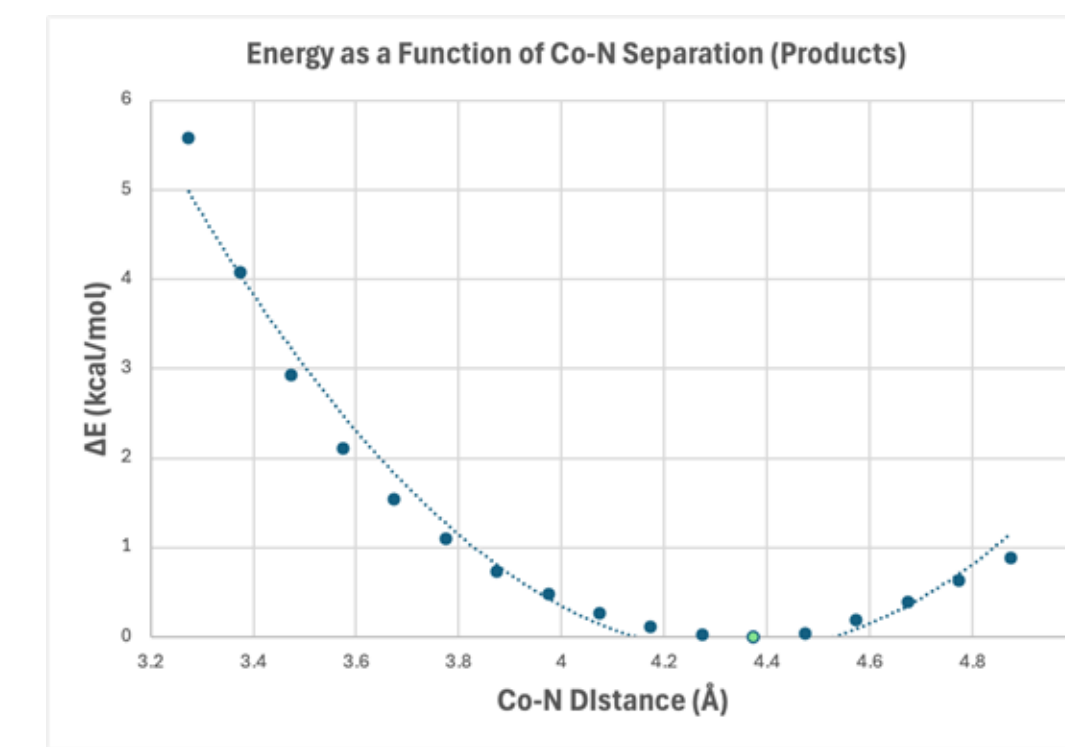
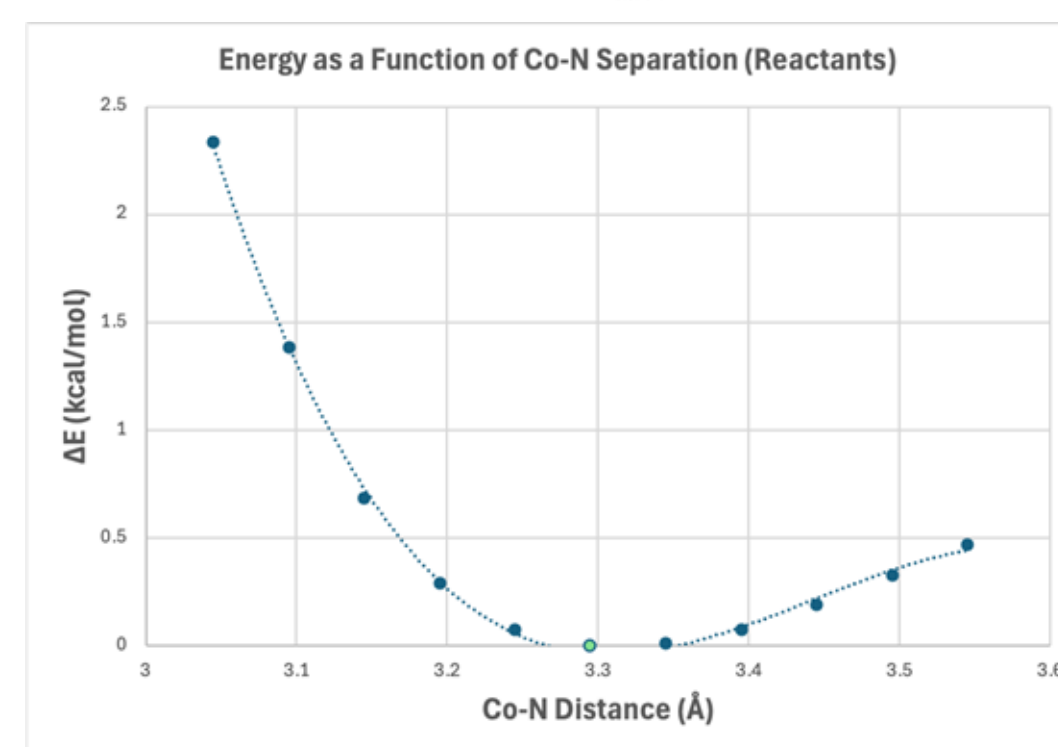
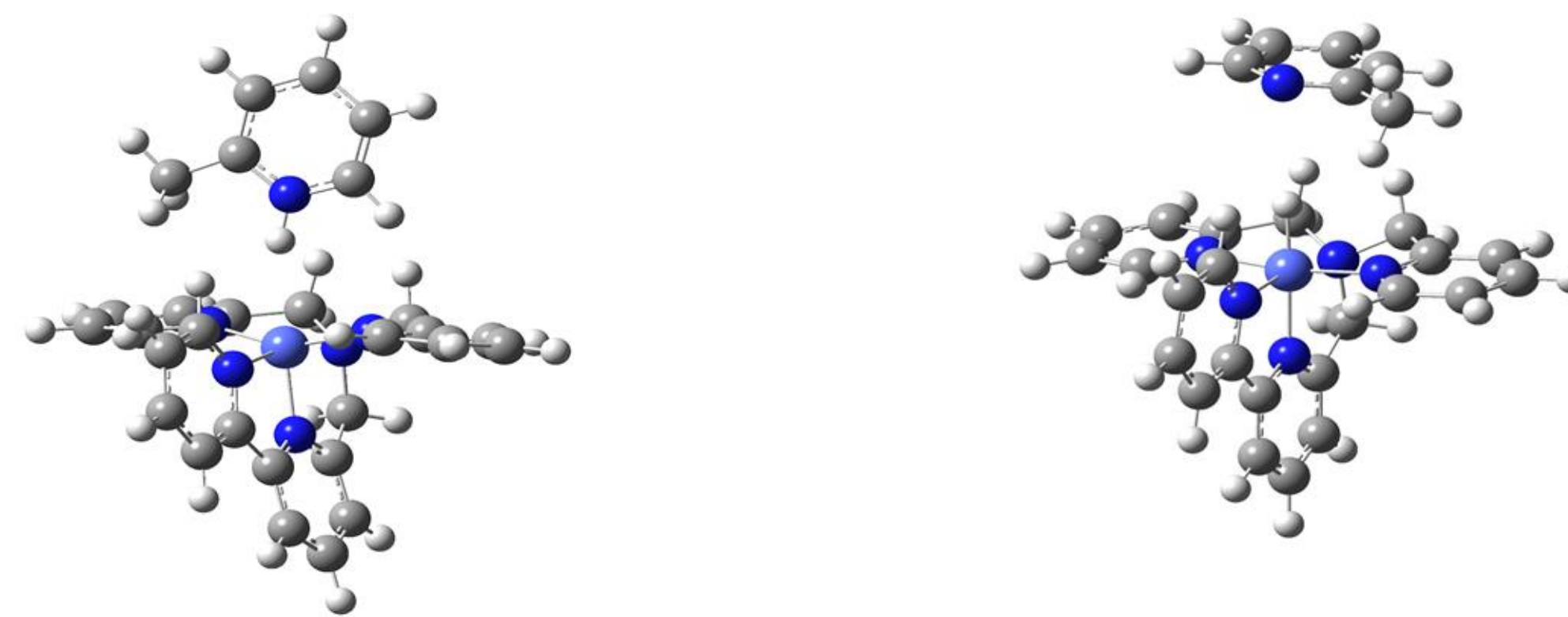
Work Term Effect



Work Terms

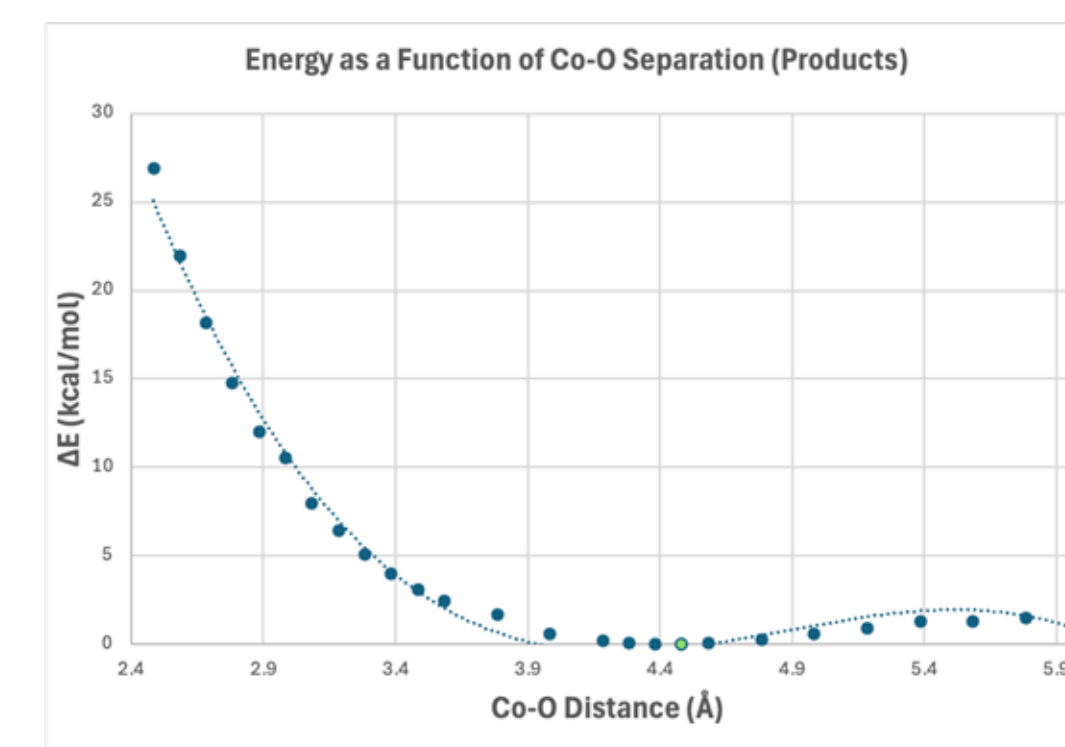
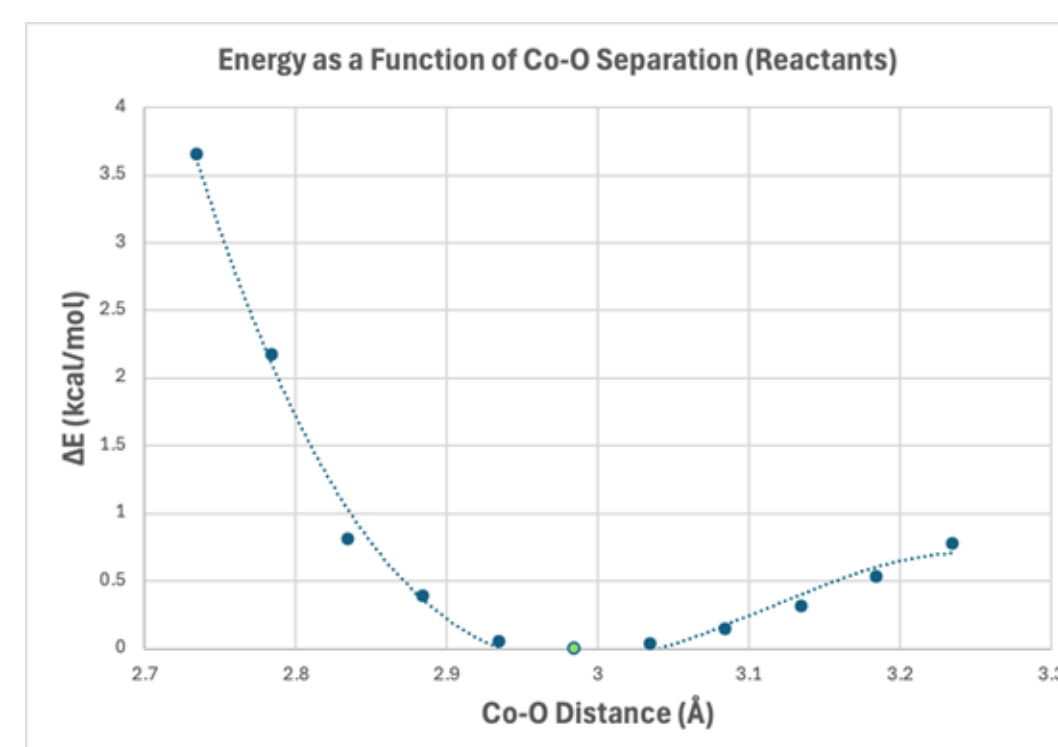
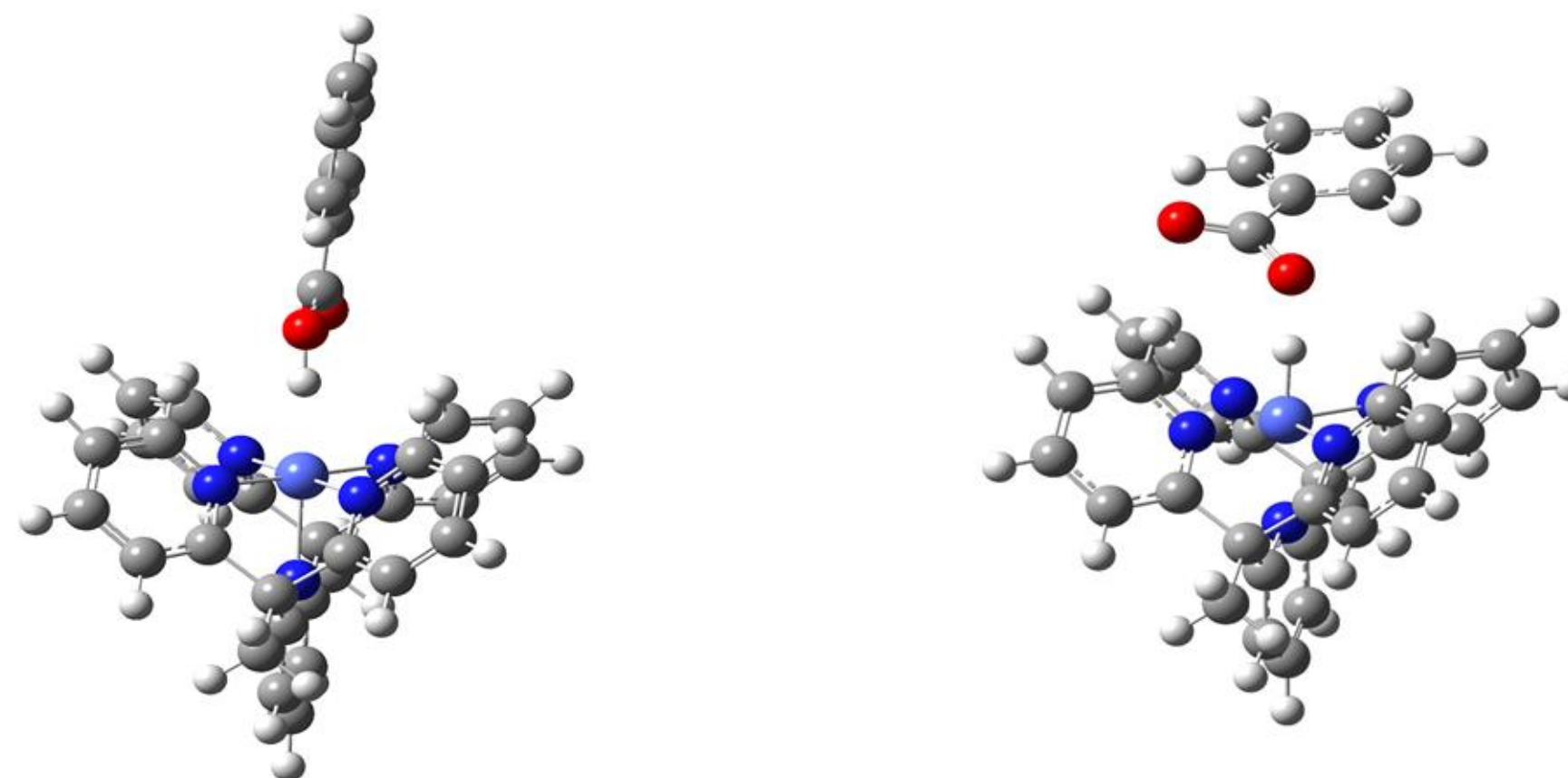
CoPy4 + 2-Methylpyridinium

Work Term for forming Reactant Donor-Acceptor Complex = 14.01 kcal/mol



CoPy5 + Benzoic Acid

Work Term for forming Reactant Donor-Acceptor Complex = 4.79 kcal/mol



Results & Discussion

Calculated ΔG°

CoPy4	ΔG° (kcal/mol)
2-Methylpyridinium	-6.92
Trifluoroacetic Acid	-13.67
Difluoroacetic Acid	-7.77
3,4-Dinitrophenol	-7.81

CoPy5	ΔG° (kcal/mol)
Benzoic Acid	-8.14
4-Chlorophenol	-5.06
Acetic Acid	-5.13

Reorganization Energies (λ)

CoPy4	λ (kcal/mol)
2-Methylpyridinium	8.16
Trifluoroacetic Acid	18.1
Difluoroacetic Acid	16.7

CoPy5	λ (kcal/mol)
Benzoic Acid	4.56
4-Chlorophenol	10.5

Conclusion

- PT to CoPy4 is intrinsically slower due to the large changes in metal-ligand angles:
 - o Energy associated with ligand reorganization increases reactant work term rather than reorganization energy
- For CoPy4 the work term is the dominant factor for the activation energy
- Work terms and reorganization energy are small for CoPy5 -> reaction occurs too quickly for rate constant to be measured

Future Work

- Calculate the product work term at PT distance.
- Calculate outer-sphere reorganization energies.
- Calculate transmission coefficient (which describes probability of proton tunneling).

References

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- DFT calculations were carried out using B3LYP functional and Def2TZVP basis set